NASA Langley Research Center

X-Ray systems Service Contract Agreement - Statement of Work

3/28/2026

1.0 Introduction and Background

1.1 NASA LaRC Nondestructive Evaluation sciences branch (NESB) operates a Large Volume Computed Tomographic (LVCT) system used for non-destructive evaluation of large aerospace components. To accommodate Mars Sample Return Earth Entry Vehicle (MSR-EEV) the system requires structural modification.

The objective of this contract is for Nikon to design, draft engineering drawings, and fabricate the necessary hardware components to raise both the X-ray source and the detector of the LVCT system by approximately six (6) inches.

1.2 SCOPE OF WORK

The Contractor (Nikon) shall provide all personnel, equipment, supplies, facilities, transportation, tools, materials, and supervision necessary to perform the following tasks:

Task 1: Engineering and Design

- Assess the current configuration of the LVCT system's source and detector mounts.
- Develop mechanical engineering drawings for the structural components required to elevate the source and detector by 6 inches.
- Ensure the design maintains structural rigidity, vibration dampening, and precise optical/mechanical alignment required for optimal LVCT system performance.
- Submit draft engineering drawings to the NASA Technical Point of Contact (TPOC) for review and approval prior to fabrication.

Task 2: Fabrication

- Upon written approval of the engineering drawings from the NASA TPOC, fabricate all required components.
- Ensure all materials used meet the appropriate specifications for strength, durability, and compatibility with the existing LVCT system.
- Perform quality assurance checks on the fabricated components to verify dimensional accuracy and alignment tolerances.

Task 3: Delivery and Installation/Calibration

- Deliver the fabricated components to LaRC NESB.
- Provide all associated final engineering drawings and material certifications.
- Install the components on the LVCT system, perform system alignment, and conduct verification testing to ensure the system operates within original baseline specifications at the new height.

3. SECURITY CLASSIFICATION

Unclassified

4. GLOSSARY

NDE- Non-Destructive

QA&IP- Quality Assurance and Inspection

SOW- Statement of Work

COR – Contracting Officer Representative CO – Contracting Officer

TR – Technical Representative

5. PLACE OF PERFORMANCE:

Work shall be performed at Langley Research Center, Hampton, VA

6. PERIOD OF PERFORMNACE:

The period of performance will be from the date the task issuance for a period of 6 months.

7. NASA TECHNICAL REPRESENATIVE (TR):

The TR will serve as the task technical liaison between the Contractor and the COR and will assist the COR in performing his/her duties.